Plant Systematics and Evolution (2026) 312:13
https://doi.org/10.1007/s00606-026-01984-3

RESEARCH

Croton awaensis (Euphorbiaceae), a new species of dragon's blood tree from the Chocó-Andean region

Michelle Suntaxi-Oña[1] · Catalina Quintana[1] · Daniel Bellido-Field[2] · José Murillo-A[3] · Ricarda Riina[4]

Received: 5 September 2025 / Accepted: 11 January 2026 / Published online: 8 March 2026
© The Author(s) 2026

Abstract

We describe and illustrate *Croton awaensis*, a new species of dragon's blood tree (*Croton* sect. *Cyclostigma*) with medicinal value distributed in the Chocó and Andean regions of Ecuador and Colombia. The species is a medium-sized tree occurring in humid lowland and lower montane forests, between 300 to 1500 m. It is characterized by a sublepidote indumentum, falciform stipules, 2 acropetiolar nectaries on the abaxial side of the petiole and 0–8 on the adaxial side, shortly pedicellate fruits, and bifid styles. *Croton awaensis* could be potentially confused with *C. floccosus* or *C. coriaceus*, which also grow in the northern Andean humid montane forests of Ecuador but at higher elevations. The new species grows sympatrically, sharing the same habitat and location, with *C. chocoanus*, another tree species that belongs to a different section in the genus (*C.* sect. *Luntia*). Phylogenetic analysis of DNA sequence data from the nuclear ribosomal ITS region supports the placement of the *C. awaensis* in *C.* sect. *Cyclostigma* and its closest relationship with a clade including several dragon's blood species from northern South America and Mesoamerica. We present and illustrate a set of morphological characters to aid in the identification of specimens of *C. awaensis* using images from herbarium material and field observations. A preliminary conservation assessment indicates that the new species is likely endangered due to habitat loss resulting from agricultural expansion and the scarcity of protected areas within its distribution range.

Keywords Andes · Chocó bioregion · *Croton* sect. *Cyclostigma* · Neotropics · Taxonomy

Introduction

While working on an inventory of *Croton* trees from Andean region, we came across several collections from Ecuador and Colombia that had been previously misidentified or included as indeterminate material in herbaria. A preliminary analysis of the morphology of these specimens allowed us to hypothesize that they likely belonged to an undescribed species of *Croton* sect. *Cyclostigma*. The majority of the Andean *Croton* trees belong to this section, which is among the most species-rich Neotropical clades in the genus (Riina et al. 2009; Van Ee et al. 2011). Members of sect. *Cyclostigma* are commonly known as dragon's blood trees ("sangre de drago" or "sangre de grado" in Spanish) due to the presence of copious red latex, which is widely used for medicinal purposes in several Latin American countries (Meza 1999; Jones 2003; King et al. 2020).

Croton sect. *Cyclostigma* as defined in the current classification of the genus (Van Ee et al. 2011) comprises ca. 50 species of small- and medium-sized trees. This number includes the species here described and six more species

Handling Editor: Karol Marhold.

✉ Catalina Quintana
cquintanam@puce.edu.ec

✉ Ricarda Riina
rriina@rjb.csic.es

[1] Facultad de Ciencias Exactas, Naturales y Ambientales, Pontificia Universidad Católica del Ecuador, Av. 12 de Octubre, 1076 Quito, Ecuador

[2] Escuela Superior de Ciencias Experimentales y Tecnología, Universidad Rey Juan Carlos, Calle Tulipán S/N, 28933 Móstoles, Spain

[3] Instituto de Ciencias Naturales, Universidad Nacional de Colombia, Cr. 30, 45-03 Bogotá, Colombia

[4] Real Jardín Botánico, CSIC, Plaza de Murillo 2, 28014 Madrid, Spain

described in recent years (Feio et al. 2018a; Farías et al. 2019; Mateo-Ramírez & Riina 2020). Although the section has a wide geographic range, stretching from Mexico to northern Argentina, most of its diversity is concentrated in the montane forests along the Andes with ca. 32 species. The second center of diversity is located in the Brazilian Atlantic forest, which is home to approximately ten species (Santos et al. 2017; Farias et al. 2019).

In this paper we name, describe, and illustrate a new species of the dragon's blood clade of *Croton* (*C.* sect. *Cyclostigma*). We compare the morphology and distribution of *C. awaensis* sp. nov. with those of *C. floccosus* and *C. coriaceus*, the two morphologically most similar species within section *Cyclostigma*. The new species is also placed in a phylogenetic context based on sequence data from the ITS region (nuclear ribosomal Internal Transcribed Spacers) to confirm its placement in sect. *Cyclostigma* and to infer its phylogenetic relationships. Finally, we provide a distribution map and a preliminary assessment of the species conservation status.

Materials and methods

Phylogenetic analysis

We used the ITS region, which has been employed extensively in numerous studies of *Croton* phylogenetics (e.g., Berry et al. 2005; van Ee & Berry 2011, 2021; van Ee et al. 2011; Haber et al. 2017; Riina et al. 2021; Chaves-Beyoda 2022; Rossine et al. 2025), as a DNA-barcode with reasonable results despite its limitations (Feliner & Rosello, 2007). Taxon sampling for phylogenetic analysis included 37 accessions corresponding to 34 species of *Croton* and one outgroup (*Brasiliocroton mamonhinha*), which is well-established as the sister-group to the entire genus (Berry et al. 2005; Riina et al. 2014). Of these, 19 sequences belong to the focal group (*C.* sect. *Cyclostigma*) including two newly generated sequences of *C. awaensis*, representing its distribution in Ecuador and Colombia. Most of the sequences used were from previous phylogenetic analyses at the genus level (van Ee et al. 2011) or from studies with a particular focus on sect. *Cyclostigma* (Riina et al. 2009; Mateo-Ramírez & Riina 2020), and were downloaded from GenBank (https://www.ncbi.nlm.nih.gov/genbank/). See Table 1 for GenBank accession numbers and voucher information.

The DNA extraction of two specimens of *C. awaensis*, amplification, sequencing, and alignment followed the same laboratory and bioinformatics procedures used in Rossine et al. (2025). The aligned ITS matrix is provided as a fasta file (Online Resource 1). The phylogenetic analysis was performed in MrBayes ver. 3.2.7 (Ronquist et al. 2012) applying the GTR + I + G model following Masa-Iranzo et al. (2021). We did 2 runs for 10 million generations sampling every 1000th generation, and assessed the standard deviation of split frequencies for convergence (< 0.01). Tracer ver. 1.7.2 (Rambaut et al. 2014) was used to evaluate the Markov chains. The first 25% of MCMC samples were discarded as burn-in. Using the sumt command we obtained a Bayesian tree with posterior probability (PP) values by computing a majority rule consensus of the post-burn-in trees combined from both runs. We used FigTree ver. 1.4.4 (Rambaut 2006) to visualize and edit the consensus tree. We also performed a Maximum Likelihood analysis in IQ-TREE version 1.6.12 (Lam-Tung et al. 2015), using the same model as in the Bayesian analysis.

Morphological analysis and taxonomy

We examined collections of the putative new species and compared them with similar species of *Croton*. From this initial assessment the morphological comparison was narrowed down to the most similar species in *C.* sect. *Cyclostigma* (*C. coriaceus*, *C. floccosus*), which also occur near the area of distribution of the new species. We studied specimens of these three species from the following herbaria: AAU, B, CHEP, COL, CUVC, DAV, ECUAMZ, F, G, GB, GH, HUTI, K, LE, LOJA, MA, MEXU, MICH, MO, NY, PE, PH, PSO, Q, QAP, QCA, QCNE, SIO, TEX, U, US, and WIS. The protologues as well as the type materials and additional specimens from morphologically similar species in *C.* sect. *Cyclostigma* were also examined (Online Resource 2). Acronyms of herbaria follow Index Herbariorum (https:// sweetgum.nybg.org/science/ih/). For morphological descriptions we used a stereomicroscope with $10 - 60\times$ magnification. The morphological descriptions follow recent taxonomic and morpho-anatomical studies in *Croton* (Vitarelli et al. 2015, 2021; Feio et al. 2018a, 2018b; Riina et al. 2021). For trichome terminology, we followed the most recent classification of *Croton* trichomes (Pinto-Silva et al. 2023).

Compilation of data for conservation evaluation

Field trips were carried out during 2025 to verify some of the known localities of the new species, search for new populations, and to observe specific characters in vivo. The information about habitat, geographic coordinates, and locality was taken from examined herbarium collections and direct field observations. We converted the sexagesimal coordinates included in herbarium labels to decimal format to allow for proper recognition and processing by spatial analysis tools such as ArcGIS, and GeoCAT. For those without coordinates, we estimated them with Google Earth using locality and habitat information provided in

Table 1 Specimen voucher information and GenBank accession numbers of the nuclear ribosomal ITS sequences used in the phylogenetic analysis

Species	Accession	Voucher	Country
Brasiliocroton mamoninha P.E.Berry & Cordeiro	EU586944	*Pirani 4947* (SPF)	Brazil
C. arboreus Millsp.	EU478029	*van Ee 472* (WIS)	Mexico
C. awaensis sp. nov.	PX605297	*Patiño 112* (COL)	Colombia
C. awaensis sp. nov.	PX605298	*Riina 1416* (MA)	Ecuador
C. beetlei Croizat	EU586916	*Riina 1512* (WIS)	Bolivia
C. calcareus Riina & Mateo-Ram.	MT106663	*Hampshire 1161* (MA)	Mexico
C. caracasanus Pittier	DQ227525	*Riina 1288* (WIS)	Venezuela
C. conduplicatus Kunth	EU477957	*Lucena 598* (WIS)	Brazil
C. coriaceus Kunth	EU586921	*Riina 1403* (WIS)	Ecuador
C. cuneatus Klotzsch	EU497735	*Riina 1491* (MICH)	Peru
C. cupreatus Croizat	EU586919	*Riina 1408* (WIS)	Ecuador
C. curiosus Croizat	EU586906	*Zuloaga 8438* (SI)	Argentina
C. draco Schltdl. & Cham.	EU478006	*van Ee 347* (WIS)	Costa Rica
C. echinocarpus Baill.	EU586922	*Riina 1371* (WIS)	Brazil
C. eichleri Müll.Arg.	EU586949	*Riina 1525* (WIS)	Brazil
C. floccosus B.A.Sm.	AY971210	*Berry 7610* (WIS)	Ecuador
C. floccosus B.A.Sm.	EU586923	*Ceron 6909* (QCNE)	Ecuador
C. gossypiifolius Vahl	EU586924	*Mota s.n.* (WIS)	Venezuela
C. goudotii Baill.	EU586946	*Hoffmann 248* (K)	Madagascar
C. gratissimus Burch	AY971260	*Zimba 901* (MO)	Zambia
C. hibiscifolius Kunth ex Spreng.	EU586925	*Riina 1413* (WIS)	Ecuador
C. huberi Steyerm.	AY971217	*Berry 7679* (WIS)	Venezuela
C. macrobothrys Baill.	EU586928	*Riina 1522* (WIS)	Brazil
C. mutisianus Kunth (as *C. magdalenensis* in GenBank)	EU586929	*Arias 211* (NY)	Colombia
C. perspeciosus Croizat	EU586931	*Riina 1435* (WIS)	Peru
C. pilulifer Rusby	EU586932	*Riina 1500* (WIS)	Bolivia
C. priscus Croizat	EU586950	*Riina 1535* (WIS)	Brazil
C. pungens Jacq.	AY971241	*Riina 1272* (WIS)	Venezuela
C. purdiei Müll.Arg.	EU586934	*Becerra 20* (NY)	Colombia
C. rimbachii Croizat	EU586936	*Riina 1422* (WIS)	Ecuador
C. sampatik Müll.Arg.	EF421792	*Riina 1447* (WIS)	Peru
C. sapiifolius Müll.Arg.	EF421754	*Lima da Paixao 667* (CEPEC)	Brazil
C. schiedeanus Schltdl.	AY971246	*Aguilar 886* (MO)	Costa Rica
C. speciosus Müll.Arg.	AY971251	*Berry 7590* (WIS)	Venezuela
C. suberosus Kunth	EU477979	*Steinmann 1658* (WIS)	Mexico
C. urucurana Baill.	EU586937	*Riina 1371* (WIS)	Brazil
C. yavitensis Croizat	EU586918	*Beck 5710* (LPB)	Bolivia

the collection labels (Online Resource 2). The distribution map was created using ArGis online (ESRI, Inc.). We conducted a preliminary assessment of the conservation status of the new species following the IUCN criteria (IUCN 2022). We estimated the area of occupancy (AOO) and extent of occurrence (EOO) using the GeoCAT software (Bachman et al. 2011), a geospatial tool designed to support Red List threat assessments.

Results

Phylogenetic placement of the new species

The aligned ITS matrix consisted of 676 positions. The phylogenetic analysis based on Bayesian Inference and Maximum likelihood yielded identical results, so we selected the Bayesian results (Fig. 1) to describe the

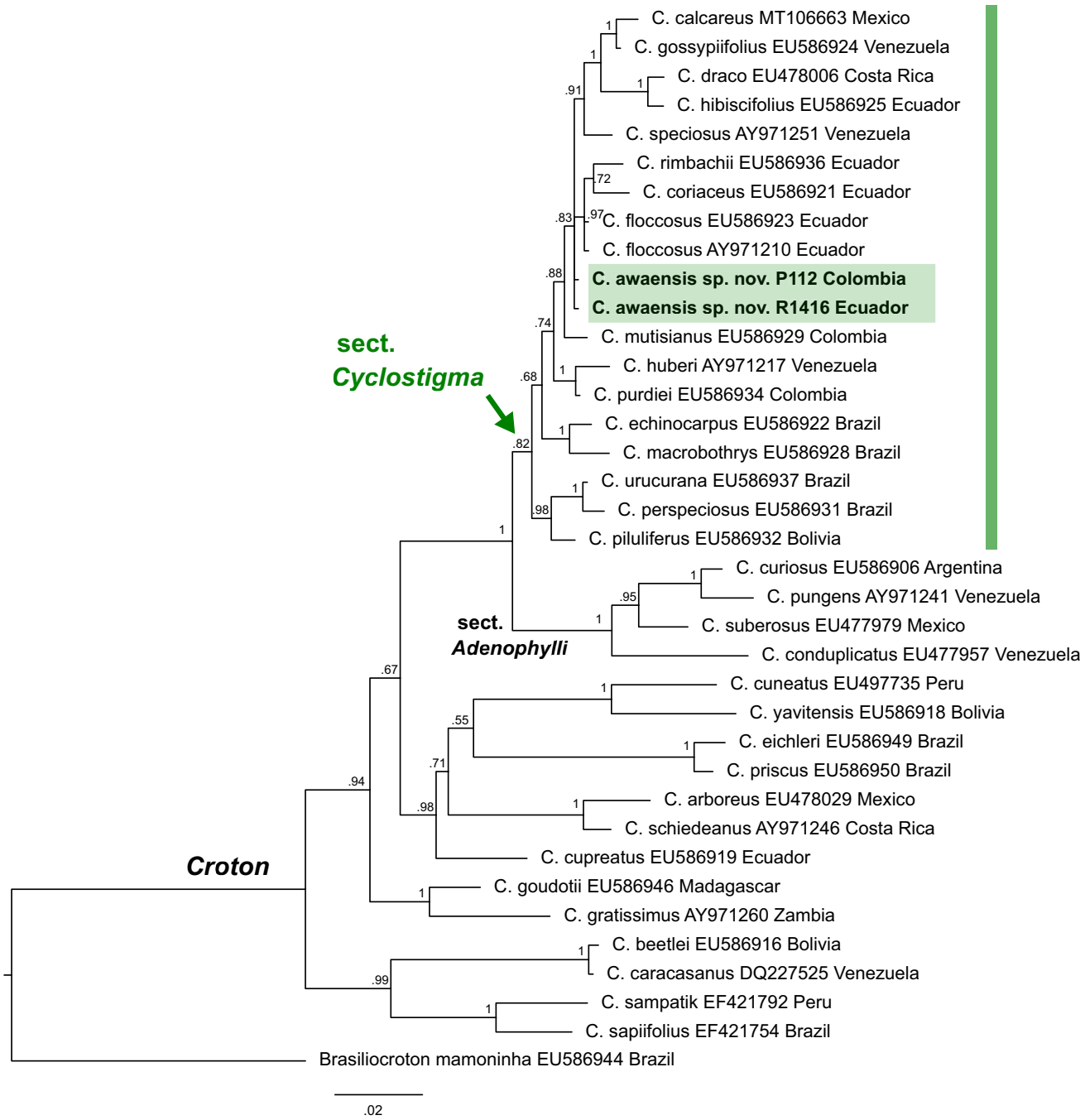


Fig. 1 Phylogeny based on Bayesian analysis of nuclear ITS sequence data from a selection of *Croton* taxa including most of the published sequences of *Croton* sect. *Cyclostigma*, two accessions of the new species *Croton awaensis* (highlighted) and *Brasiliocroton* as an outgroup. Values at nodes are Bayesian posterior probabilities. GenBank accession numbers are indicated along with the origin (country) of the accession. P112 = *Patiño 112* (COL); R1416 = *Riina 1416* (MA)

main phylogenetic patterns. All *Croton* species sampled formed a monophyletic group (Posterior Probability, PP = 1, Fig. 1) composed of two major clades with high support (PP = 0.99, 0.94). All sampled members of sect. *Cyclostigma*, including the two accessions of *C. awaensis*, were recovered as a monophyletic group with relatively moderate support (PP = 0.82) and sister to a clade with representatives from *C.* sect. *Adenophylli* (PP = 1). The two accessions of the new species were recovered within sect. *Cyclostigma* clade as part of an unresolved polytomy with relatively moderate clade support (PP = 0.83). This clade also included two well supported subclades: one

formed by *C. coriaceus*, *C. floccosus* and *C. rimbachii* (PP = 0.97), all occurring in Ecuador, and the other subclade contained species from Mexico, Venezuela and Colombia (*C. draco*, *C. gossypiifolius*, *C. hibiscifolius*, and *C. speciosus*). Sister to this polytomy was an accession of *C. mutisianus* (PP = 0.88), a species from the high Andean forests of Colombia (Fig. 1). The two sequences of *C. awaensis* were almost identical, differing in a single position (G vs. R), however they were not recovered as monophyletic.

Morphological analysis and taxonomy

The analysis of morphological characters and ecological and distribution data allowed us to propose a new taxon in sect. *Cyclostigma* (Figs. 2, 3 and 4). The formal description and additional data on the new species are included in the taxonomic treatment below. The analysis of the key characters of section *Cyclostigma* revealed marked differences between *C. awaensis* and its morphologically closest species in the section (Table 2, Fig. 5). The most useful characters to separate the *C. awaensis* from its most closely related species are summarized in Table 2 and further discussed below.

Species conservation status evaluation

Based on the GeoCAT analysis, we obtained an EOO of 6,266 km^2, indicating that *C. awaensis* could be Vulnerable (VU), and an AOO of 64 km^2, suggesting it should be classified as Endangered (EN), according to IUCN Criterion B. In addition, the new species was preliminary assessed as Endangered under Criterion B2 because it has a restricted Area of Occupancy (AOO) of 60 km^2, which is below the threshold for Endangered (< 500 km^2). *Croton awaensis* is known from a limited number of localities, and there is evidence of a continuing decline in the quality and extent of its natural habitat. These factors meet subcriterion (a)—restricted number of locations—and subcriterion (b)(iii)—continuing decline in habitat quality. Even if the Extent of Occurrence (EOO) meets the threshold for Vulnerable, the more severe AOO-based category applies. Therefore, the conservation status of *C. awaensis* was provisionally assigned as EN B2ab(iii).

Discussion

Species discovery in tropical areas remains a work in progress for many plant groups, particularly in species-rich genera such as *Croton*. Documenting and describing new biodiversity in this genus are urgent tasks as many of the recently described species occur in highly threatened species-rich biomes such as the Andean forest (e.g., Feio et al. 2018a, b; Riina et al. 2021), or the Brazilian Atlantic rain forest (e.g., Caruzo et al. 2008; Farias et al. 2019) and the seasonally dry forest (e.g., Sodré & Silva 2022; Martín-Muñoz et al. 2024; Santos et al. 2024; Zapata-Correa et al. 2024). The Chocó-Darién-Western Ecuador region (Myers et al. 2000)—or the Pacific lowlands sensu Bernal et al. (2016)—is also a biodiversity hotspot where botanical knowledge is still largely incomplete. The discovery and description of *C. awaensis* contribute to our understanding of the flora of this important and threatened tropical bioregion.

Most of the records of the *C. awaensis* are from Ecuador with only two collections known from Colombia (Fig. 4). The scarcity of records in Colombia is likely a result of the limited botanical explorations in this country owing to more than five decades of armed conflict (Álvarez 2001; Sierra et al. 2017). Future botanical sampling of the Chocó-Andean region of Colombia will probably find additional occurrences of *C. awaensis* on the Colombian side. From a biogeographical perspective, the new species is interesting because it is one of two species of sect. *Cyclostigma* reaching the lowland forests of the Chocó bioregion. The other species of sect. *Cyclostigma* occurring in lowland areas of the Chocó bioregion is the Ecuadorian endemic *C. churutensis* Riina & Cornejo. However, *C. churutensis* occurs in seasonally dry forest from sea level to 700 m of elevation and its geographic range is more than 200 km apart from that of *C. awaensis* since it occupies a different latitudinal zone near the coast of southwestern Ecuador (Riina et al. 2007).

Our phylogenetic analysis shows that the topology of the sampled *Croton* taxa (Fig. 1) is congruent with previous phylogenetic studies in *Croton* (Berry et al. 2005; Van Ee et al. 2011). Likewise, the structure of sect. *Cyclostigma* clade is also consistent with analyses that included a much broader representation of this section (Riina et al. 2009; Mateo-Ramírez & Riina 2020). The ITS data corroborate our initial hypothesis (based on morphology) that the new species belongs to sect. *Cyclostigma*. However, similar to the phylogenetic pattern found in previous studies, ITS alone cannot fully resolve (i.e., with high support) species relationships within sect. *Cyclostigma*. Although the phylogenetic results do not show the two *C. awaensis* accessions as monophyletic, they do not contradict the idea that they belong to the same species. To increase resolution and provide a solid phylogenetic framework for sect. *Cyclostigma* and a better delimitation of *C. awaensis,* future research in this group should employ a multigene or a phylogenomic approach using a comprehensive sampling of the section and a large representation of the nuclear genome.

While ITS data alone do not provide a strong base for the taxonomic delimitation of *C. awaensis*, several morphological characters, along with geographic distribution, are useful for distinguishing the new species from similar species in its section. The species in sect. *Cyclostigma* that are most similar in morphology to *C. awaensis* are *C. floccosus* and *C. coriaceus*, which are also the closest in the section in


a
b
c
d
e
f
g
h
i
j
k

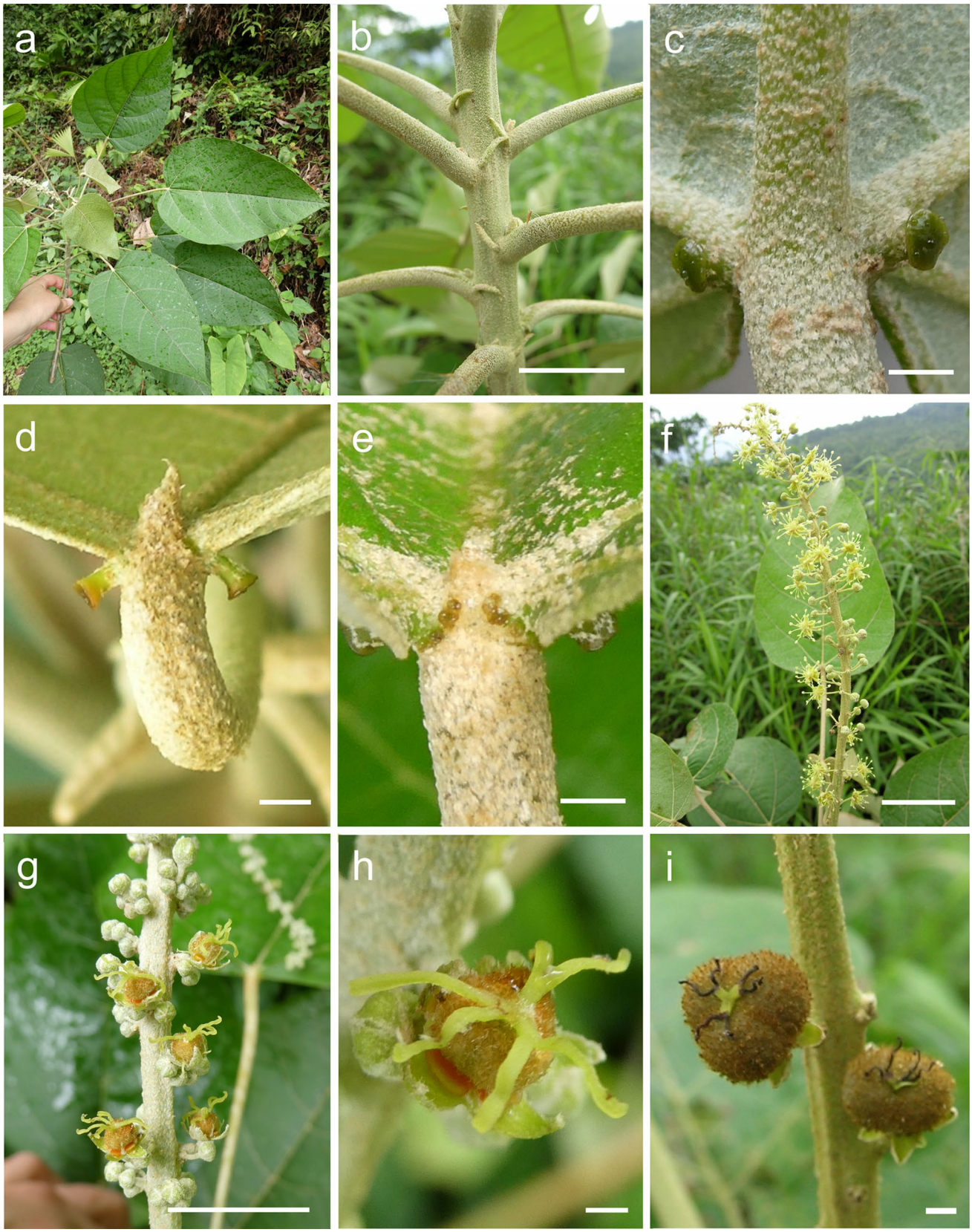
◂**Fig. 2** Images of *Croton awaensis* sp. nov. from herbarium specimens: **a** view of a fertile branch showing the characteristic discolor leaves; **b** stipules; **c** dense indument of sublepidote trichomes on the adaxial side of a young leaf; **d** sparse indumentum of sublepidote trichomes on the adaxial side of a mature leaf; **e** detail of the two abaxial acropetiolar nectaries; **f** adaxial side of a leaf base showing sessile and stipitate acropetiolar nectaries; **g** detail of four long-stipitate adaxial nectaries seen from the abaxial side of the leaf; **h** staminate cymules with flowers and buds; **i** pistillate flowers showing the six tips of the bifid styles; **j** a fruit close to maturity; **k** view of seeds from ventral side (top two showing the caruncle) and dorsal side (bottom). Vouchers: a, c-f, h *Jaramillo 23311*, QCA; b, j *Riina 1416*, MA; g, i *Barfod 41491*, QCNE; k *Grijalva 474*, QCNE. Scale bars: a (10 mm); b, h–k (2 mm); c-g (1 mm)

terms of geographic range. The three species can be distinguished by a set of morphological characters including trichome type, stipule shape, position, number and shape of acropetiolar nectary glands, as well as pistillate flowers and fruits features (Table 2, Fig. 5). Additionally, the geographic distribution and habitat of *C. awaensis* are also informative for species identification, since there is no overlap with the distributions and habitats of *C. floccosus* and *C. coriaceus* (Table 2, Fig. 4).

The only *Croton* species that occurs sympatrically with *C. awaensis* is *C. chocoanus* Croizat. They grow frequently next to each other in the Chocó forests of Ecuador (field observation) and the same probably happens in the Colombian side, where *C. chocoanus* also occurs. The two species are easy to tell apart—since they belong to two different sections in the genus. *Croton chocoanus* (sect. *Luntia*) can be distinguished by its lobed leaves, sparse laminar nectary glands on both sides of the leaf blade, and multifid styles (vs. entire leaves, absence of laminar nectaries, and bifid styles in *C. awaensis*).

Some specimens of *C. awaensis* have been annotated in herbaria with the names *C. lechleri* Müll.Arg., *C. mutisianus* Kunth or *C. sampatik* Müll.Arg. The first two species belong to sect. *Cyclostigma* and are indeed superficially similar to *C. awaensis*. *Croton lechleri* is currently under taxonomic revision, since the use of the name in the scientific literature and on herbarium collections appears to involve several confused taxa. However, even in its broad sense (*C. lechleri* s.l.) it can be differentiated from *C. awaensis* by its stellate indumentum (sublepidote in *C. awaensis*), subulate to cuculiform stipules, and its distribution in the eastern Andes-Amazonian foothills. So far *Croton mutisianus* is known only from Colombia, where it is widespread across high Andean humid montane forests between (1400)1800–2800 m; Murillo (1999) indicated the upper limit at 2400 m but our recent revisions of herbarium material shows that it can reach up to 2800 m. Besides geography and habitat, *C. mutisianus* can be differentiated from *C. awaensis* by its stellate, multiradiate, and dendritic trichomes, as well as the presence of more conspicuous and variable stipules, which go from subulate to slightly foliose with (sometimes) slightly dissected margins. The sequence of *C. mutisianus* included in our analysis (Fig. 1) shows that it is sister to the clade containing *C. awaensis*, however additional sampling with a better representation of the distribution and morphological variation of *C. mutisianus* is needed to confirm its taxonomic delimitation, phylogenetic position, and its relationship with *C. awaensis*. Finally, *C. sampatik* (sect. *Sampatik*) is different from *C. awaensis* in having abaxial, stipitate nectary glands on the lamina near the leaf margin, larger deeply lobed capsules (ca. 10 mm long), laterally compressed seeds, and a distribution in the Andes-Amazonian foothills.

Although our preliminary conservation assessment rendered *Croton awaensis* as Endangered, this result should be taken with caution. The species has a relatively wide geographic range along the western lower and middle slopes of the northern Andes of Colombia and Ecuador, but there are potential collection gaps in its distribution (Fig. 4). Most of the collection records are concentrated in the Esmeraldas and Carchi provinces in Ecuador. During our visit to the localities near the Lita–San Lorenzo road (Esmeraldas) in June 2025, we verified these occurrences and made new herbarium collections in nearby localities. We observed individuals of different ages (from juvenile to mature trees) suggesting that the species is regenerating. Specimens of *C. awaensis* have also been collected within the areas of Awá ethnic reserves in Ecuador and Colombia (Municipalities of Ricaurte and Barbacoas). Although these reserves are part of the national systems of protected areas (SNAP in Ecuador and SNIP in Colombia), they have been affected by mining and deforestation for agriculture and aquaculture. Outside of these reserves, a few collections have been made in Santo Domingo de los Tsáchilas Province in Ecuador. However, during our visit to these localities in June 2025, we were unable to find the species, as the forest vegetation has been largely converted into banana and cacao plantations. On the other hand, the closest SNAP protected area to the known distribution of *C. awaensis* is the Reserva Ecológica Cotacachi-Cayapas (RECC). It is possible that the species occurs in RECC since low montane forests are present within its limits (Elbers 2011). Other plant species with distributions and habitat preferences similar to those of *C. awaensis* are found within the limits of the RECC (e.g., Cornejo 2009; Orozco & Canal 2011; Cornejo & Mori 2012). Further botanical explorations targeting the gap areas within the current distribution range of the new species (Fig. 4) are necessary to better assess its conservation status.


a
b
c
d
e
f
g
h
i

◂**Fig. 3** Images of *Croton awaensis* sp. nov. from specimens in vivo: **a** young branch showing the glabrescent adaxial surface of mature leaves and a dense indument on young leaves; **b** detail of the stipules; **c** abaxial acropetiolar nectaries; **d** another view of the abaxial nectaries; note that this specimen lacks nectaries on the adaxial side (see next); **e** view of the adaxial leaf side showing up to eight small nectaries; **f** upper part of an inflorescence with staminate flowers; **g** lower part of an inflorescence with bisexual cymules; staminate flowers in bud; **h** pistillate flower showing the bifid styles; **i** mature fruits with persistent styles. Images from the holotype (a, c, e, g, h) and *R. Riina* and *L. López 1416* (b, d, f, i). Photos: R. Riina. Scale bars: b, f, g = 10 mm; c, d, e, h = 1 mm

Taxonomic treatment

Croton awaensis Riina & Suntaxi, **sp. nov.—** TYPE: ECUADOR, Esmeraldas, km después de Durango, en dirección Lita, 1°01′57 ′′N, 78°36′59.5′′W, 350 m a. s. l., 28 Jun 2025 (pist. fl.), *Riina R., Quintana C., Suntaxi M.* and *Alvia P. 3133* (holotype: QCA!; isotypes: HUPTL!, MA!, MO!, QCNE!) (Figs. 2, 3).

Vernacular names: Sangre de drago (in Spanish), sangrano (in Awaví) (reported in labels of herbarium specimens: *Grijalva* et al. *474*; *Narváez* et al. *1085; Patiño-Ch. and V. Pascal 112*).

Etymology: The specific epithet is given in honor of the Awá indigenous people, who primarily inhabit the area of northern Ecuador and southern Colombia (Orejuela 1999). The Awás have traditionally used several *Croton* species for medicinal and ritual purposes. According to the labels of the herbarium specimens and one literature source (Ministerio de Educación del Ecuador 2021), *Croton awaensis* is used for eczema, wound healing and fungal infections.

Description: Monoecious trees, up to 26 m high,10–40 cm dbh; young branches with a dense indumentum of sublepidote trichomes; mature branches glabrescent; latex red (main trunk and old branches) and yellowish (young branches). Stipules linear, caducous (only present on apical and young nodes), 4.9–5.2 × 0.8–1 mm, falciform, often embracing the stem, apex long-acuminate, margin with minute, scattered, globose glands (colleters), 0.1–0.2 mm long, glabrous on the abaxial side, with a dense indumentum of sublepidote trichomes on the adaxial side. Petioles 7–12.5 (–16) cm long, slightly grooved adaxially, with a dense indumentum (young petioles) to sparse indumentum (old petioles), of sublepidote trichomes; acropetiolar nectary glands 0–8 on the adaxial side, 2 on the abaxial side, patelliform, 0.4–1.4 mm diameter, sessile or stipitate, stipe 0.1–1.9 mm long. Leaf blades ovate, 8–19 × 5–16.24 cm; indumentum on the abaxial side dense, with sublepidote trichomes, glabrous on the adaxial side; base truncate, slightly rounded or cordate; apex acuminate to long-acuminate; margin entire with minute, globose glands (colleters), ca 0.2 mm diameter; venation 6–8 pinnate, brochidodromous; primary and secondary veins raised on both surfaces. Inflorescences terminal, erect, apical portion nodding, up to 35.5 cm long, axis costate, with a dense indumentum of sublepidote trichomes; bracts triangular, ca. 1 × 1 mm; cymules bisexual,each with 1 pistillate flower and 6–8 staminate flowers, toward the base of the inflorescence axis (staminate flowers at the base early caducous), regularly spaced at the lower section and more congested toward the apex. Staminate flowers pedicellate, pedicel 6–9 mm long, elongating with age; sepals 5, valvate, triangular-ovate, 2.5–3 × 0.9–1.3 mm, glabrous adaxial surface, abaxial surface with a dense indumentum of sublepidote trichomes; petals 5, narrow-elliptical, 2–2.4 × 0.9–1 mm, with scattered simple or stellate trichomes on abaxial side, margin lanate; receptacle densely pilose; stamens 15–16; filaments glabrous, 3–4 mm long, anthers 1–1.3 × 0.7–1 mm. Pistillate flowers pedicellate, pedicel 2.5–4.2 mm long, elongating with age, with sublepidote trichomes; sepals 5, valvate, triangular-ovate, 2.1–2.5 × 1.1–1.4 mm, glabrous on the adaxial side, abaxial side with a dense indumentum of sublepidote trichomes; petals absent or reduced to a filiform structure, 1.8–2.1 mm long; nectary disk segmented, segments opposite to the sepals, 1.8–2 mm long; ovary subglobose, with a dense indument of sublepidote, porrect, trichomes; styles bifid (six terminal branches), connate portion 0.9–1.1 mm long, free portion 2–2.4 mm long, glabrous, persistent. Capsules depressed subglobose, 4.1–5 × 4.5–5.3 mm, apical portion wider and slightly lobed; septicidally and loculicidally dehiscent; columella 3.3–4.5 mm long; seeds ovate, 3.5–4 × 2.9–3.1 mm, brown, coarsely rugose, shiny, caruncle triangular-trapezoid, 0.8–1 × 0.9–1 mm.

Diagnosis: *Croton awaensis* belongs to *C.* sect. *Cyclostigma*, and within this section it is most similar to *C. floccosus*. It can be distinguished from the latter by the shorter pedicel of pistillate flowers (2.5–4.2 mm vs. 11–24 mm in *C. floccosus*), smaller pistillate sepals (2.1–2.5 × 1.1–1.4 mm vs. 4.1–4.7 × 1.8–2.7 mm), and smaller fruits (4.1–5 × 4.5–5.3 mm vs. 6–7.5 × 6.4–7.7 mm).

Phenology: Data from herbarium specimens and our own field observations show that plants flower from April–June, and fruiting from November to February.

Habitats: Primary and secondary Chocó lowland and Andean forests, in areas with high annual precipitation (4000–8000 mm).

Distribution area: The new species occurs in northern Ecuador (Carchi, Esmeraldas and Santo Domingo de los Tsáchilas provinces) and southwestern Colombia (Department of Nariño), between (320–) 500–1500 m (Fig. 3; Online Resource 2). A sterile specimen (*Hernández 608*, MEDEL) located further north in Colombia (Department of Huila, Acevedo Municipality) could represent a disjunct population of *C. awaensis*, but verification

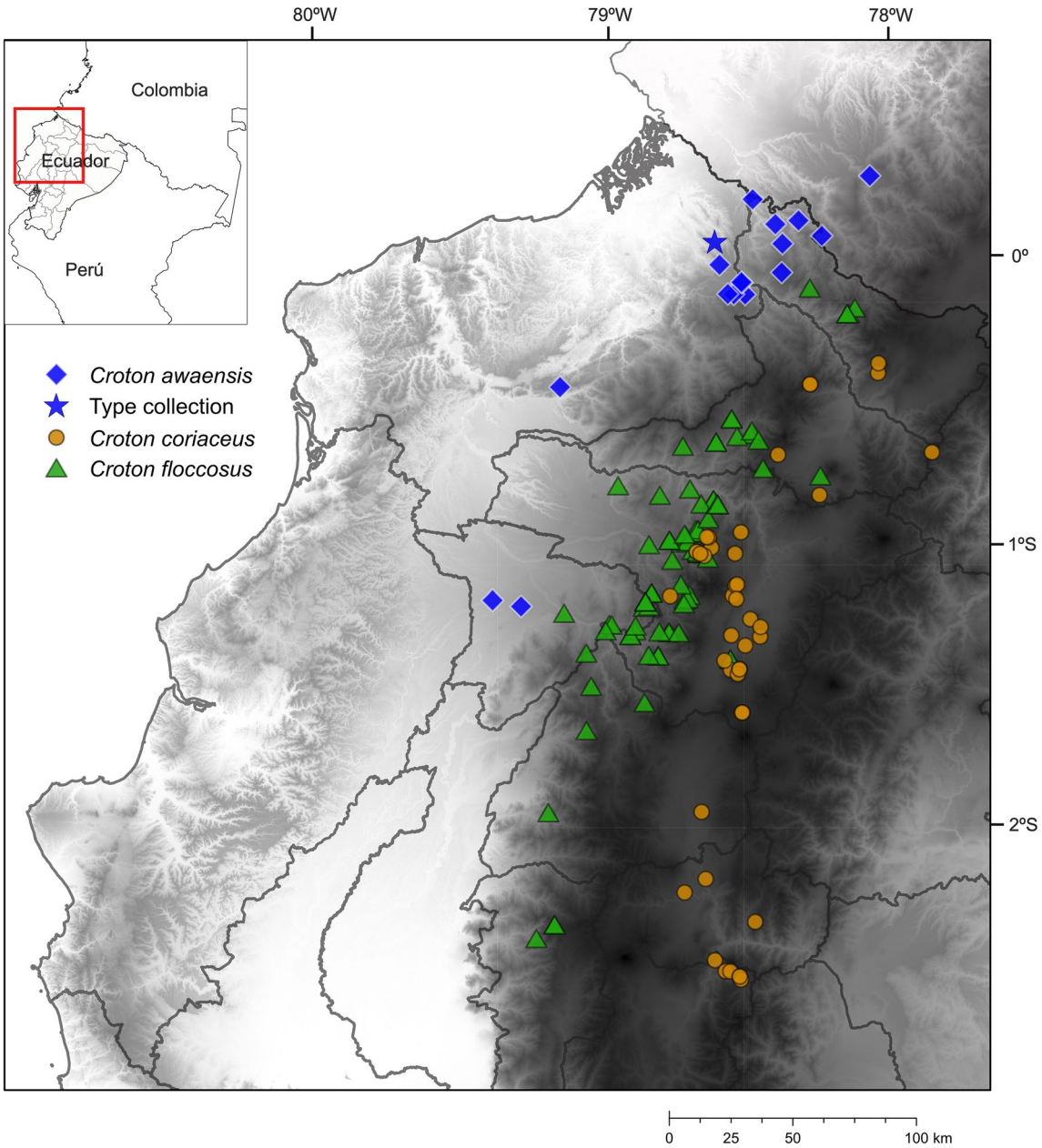


Fig. 4 Distribution map of the known records of *Croton awaensis* (blue diamonds), *C. coriaceus* (yellow circles) and *C. floccosus* (green triangles); the type locality of the new species is indicated with a star symbol. See Online Resource 2 for voucher information

of its identity is pending the availability of fertile material from this area.

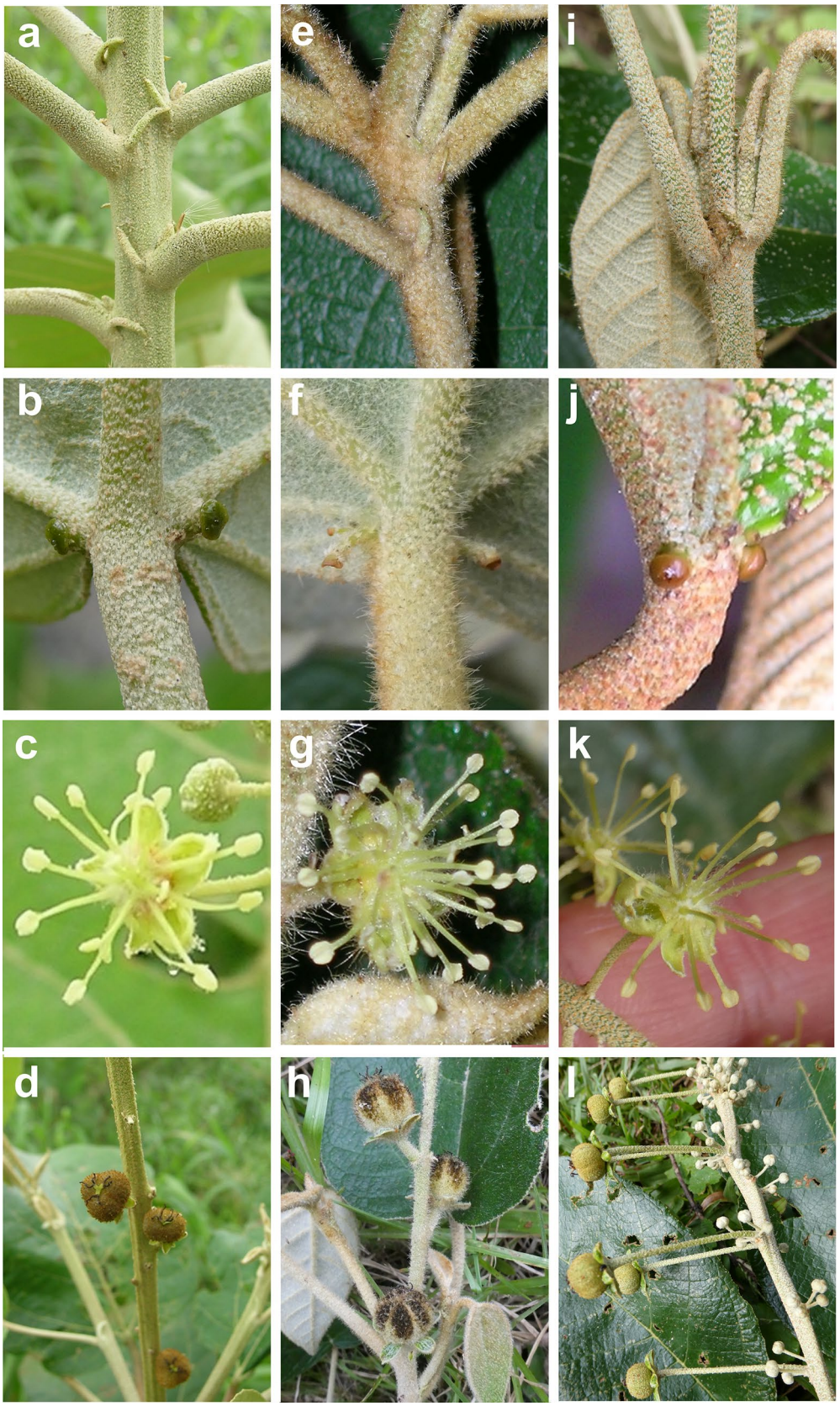
Additional specimens examined: **Colombia**. **Nariño**. Mpio. Barbacoas, Resguardo Indígena Awá Cuambí-Yaslambí, Cuambí, 1°16′57.6′′N, 78°05′36.1′′W, 550 m a. s. l., 13 Apr 2004, *A. Patiño-Ch.* and *V. Pascal 112* (COL 519233, COL 519404); Mpio. de Ricaurte, Resguardo Indígena Nulpe Medio. Andalucía, camino a Piguantís, 1°4′N, 78°15′W, 780 m a. s. l., 13 Jan 1996, *M.S. Gonzáles* and *B.R. Ramírez 1700* (PSO, QCA). **Ecuador**. **Carchi**. Maldonado, Parroquia Tobar Donoso, Reserva Etnica Awa, Sabalera, 1°10′57.1′′N, 78°29′43.5′′W, 900 m a. s. l., 22 Nov 1992, *C. Aulestia, E. Aluestia E.* and *M. Guanga 748* (MO, QCNE, WIS); San Marcos, 1°7'N, 78°20'W, 660 m a. s. l., 19 Jan 1983, *A. Barfod 41491* (AAU, QCA, QCNE); Tulcán, Reserva Etnica Awá, Parroquia El chical, Sector Gualpí Medio, Río Cumumbí, 1°02′00.0′′N 78°23′07.5′′W, 1150 m a. s. l., 19 Feb 1993, *A. Grijalva, C. Aulestia* and *J. Taicuz 474* (MO, LOJA, QCNE); Tulcán Cantón, Reserva Indígena Awá, Comunidad San Marcos, 25 km al NW de El Chical, Parroquia Maldonado, 1°06′00.0′′N, 78°24′46.7′′W, 1500 m a. s. l., 16 Nov

Table 2 Comparison of *Croton awaensis* sp. nov. with the two morphologically most similar species occurring near its geographic range

Character	*C. awaensis* sp. nov.	*C. coriaceus*	*C. floccosus*
Leaf	Membranaceous	Coriaceus	Coriaceus
Trichomes on abaxial side of leaf blade	Sublepidote	Multiradiate estipitate or stellate stipitate	Stellate porrect
Trichomes on adaxial side of leaf blade	Sublepidote	Multiradiate, porrect	Sublepidote, rarely sublepidote porrect
Acropetiolar nectary glands	0–8 on the adaxial side, 2 on the abaxial side	0 on the adaxial side, 2 on the abaxial side	2–6 (10) on the adaxial side, 2 on the abaxial side
Stipule shape, length (mm)	Linear, falciform and embracing the stem when mature, 4.9–5.2 × 0.8–1	Linear, slightly falciform, 4.5–6 × 0.7–1	Lanceolate to linear, 4–7(–11)
Stamens	15–16, filaments glabrous	23, filaments glabrous	15–16, filaments pubescent
Pedicel length of pistillate flowers (mm)	2.5–4.2	3–10	11– 24
Sepals of pistillate flowers (mm)	2.1–2.5 × 1.1–1.4	3.5–7 × 1.1–4	4.1–4.7 × 1.8–2.7
Styles	Bifid (6 tips)	Multifid (18 tips)	Bifid (6 tips)
Fruit shape	Depressed, subglobose, apical portion wider and slightly lobed	Oblongoid	Globose
Fruit size (mm)	4.1–5 × 4.5–5.3	11–13.9 × 9–14.5	6–7.5 × 6.4–7.7
Seeds size (mm)	3.5–4 × 2.9–3.1	6.6–9.5 × 4.5–5.5	4–5.1(–7) × 3.2–5.0
Caruncle (mm)	0.8–1 × 0.9–1	1.1–1.8 × 2– 2.3	0.9–1.2 × 1.3–1.9
Elevation (m)	320–1500	1480–3900	(600–)1150–2400(–3300)
Habitat	Chocó humid forest, Low Andean humid montane forest	High Andean humid montane forest	High Andean humid montane forest
Geographic range	Ecuador (Carchi, Esmeralda, Santo Domingo de los Tsáchilas), Colombia (Nariño)	Ecuador (Carchi, Chimborazo, Cotopaxi, Imbabura, Pichincha, Tungurahua)	Ecuador (Bolivar, Carchi, Cotopaxi, Imbabura, Pichincha, Santo Domingo de los Tsáchilas)

1990, *D. Rubio, C. Quelal* and *J. Pai 1049* (MO, QCNE, DAV). **Esmeraldas**. Carretera Lita-Rio Verde, 0°57′N, 78°36′W, 320–600 m a. s. l., 13 Apr 2002, *J. Jaramillo 23311* (QCA); Quininde, Parroquia Malimpia, Cristóbal Colón, Cooperativa Tesoro Escondido, Camino al Rio Gualpi, 0°30′N, 79°8′W, 500 m a. s. l., 12 Mar 2004, *E. Narváez, L. Lewinsohn, A. Moreira* and *N. Paredes 1085* (MICH, MO, QCNE); San Lorenzo, vía a Lita, 00°56'N, 78°23'W, 600 m a. s. l., 1 Apr 1994, *W. Palacios 12227* (MO, QCA); 10 km después de Lita en la vía a San Lorenzo, 00°53′28′′N, 78°31′08′′W, 750 m a. s. l., 17 Apr 2004 (fl, fr), *R. Riina* and *L. López 1416* (MA, QCA, QCNE); entre Alto Tambo y Chuchuvi, cerca un 1,5 km antes de Chuchvi, 0°53′35.4′′N, 78°31′16.2"W, 770 m a. s. l., 28 Jun 2025, *Riina R., Quintana C., Suntaxi M., Alvia P. 3136* (MA, QAC, QCNE); San Lorenzo Cantón, 9 km NW of Lita on road from Lita to Alto Tambo or Durango?, 0°51′02.7′′N, 78°30′22.4′′W, 700 m a. s. l., 27 Jun 1994, *B. Smith 212* (DAV); San Lorenzo Cantón, 13.4 km NW of Lita on road from Lita to Alto Tambo or Durango?, 0°51′00.7′′N, 78°32′45.8′′W, 700 m a. s. l., 27 Jun 1994, *B. Smith 213* (DAV); San Lorenzo Cantón, 15.7 km NW of Lita on road from Lita to Alto Tambor or Durango?, 0°51′N, 78°34′W, 790 m a. s. l., 27 Jun 1994, *B. Smith 214* (DAV). **Santo Domingo de los Tsáchilas.** Santo Domingo de los Colorados, 00°16'S, 079°14′W, 600 m a. s. l., 18 Feb 1989, *C.E. Cerón* and *G. Benavides 6188* (MO, NY, QAP, QCNE); Santo Domingo de los Colorados, Comunidad Tsachila Otongo-Mapali, 00°15'S, 79°20'W, 500 m a. s. l., 8 Aug 1995, *E. Cerón, C. Montalvo* and *M. Chango 29188* (QAP).

Proposed Conservation Status: The conservation status of *C. awaensis* is provisionally assigned as Endangered, EN B2ab(iii). See the Results and Discussion sections for further details.


a
e
i
b
f
j
c
g
k
d
h
l

◂**Fig. 5** Images of *Croton awaensis* (**a**–**d**), *C. coriaceus* (**e**–**h**) and *C. floccosus* (**i**–**l**) showing the main distinguishing characters: **a, e, i** view of indumentum and stipules on young branches of *C. awaensis*, *C. coriaceus* and *C. floccosus*, respectively (same species order in the next items); **b, f, j** acropetiolar nectaries attached to the abaxial side of the leaf; **c, g, k** staminate flowers; **d, h, l** mature fruits showing differences in size and pedicel length. Scale: within each row (same character) all images are approximate at the same scale

Supplementary Information The online version contains supplementary material available at https://doi.org/10.1007/s00606-026-01984-3.

Acknowledgements We are grateful to the staff of all the herbaria cited above (AAU, COL, HUPTL, MA, MEDEL, MICH, MO, PSO, QCA, QPA, QCNE, WIS), for allowing us to study their collections. We thank Mayra Nacimba (QCA Herbarium), Carlos Cerón (QAP Herbarium), and Laura Clavijo (COL Herbarium) for their valuable assistance during our herbarium work. Pablo Gallego (MEDEL Herbarium) and Aida Baca (PSO Herbarium) kindly sent us images of herbarium specimens of the new species. We are also grateful to Renato Valencia and Pablo Alvia for their support during the fieldwork in Ecuador and to the Developmental Biology Lab at the Pontificia Universidad Católica del Ecuador (PUCE) for giving us access to their photographic equipment. We acknowledge the Molecular Systematics Laboratory of the Institute of Natural Sciences, National University of Colombia, for their assistance with the sample of the new species from COL. We also acknowledge the staff of the Molecular Systematics Laboratory of the Real Jardín Botánico-CSIC for their continuous support to our project.

Author contributions M. S.-O. Methodology, Formal Analysis, Investigation, Writing original draft. C.Q. Conceptualisation, Methodology, Formal Analysis, Investigation, Writing original draft, Supervision. D. B-F. Methodology, Formal Analysis, Investigation. J.M.A. Investigation, Writing-editing. R.R. Conceptualisation, Methodology, Formal Analysis, Investigation, Writing original draft, Supervision, Project Administration, Funding Acquisition.

Funding Open Access funding provided thanks to the CRUE-CSIC agreement with Springer Nature. This work was supported by the DRAGORED project funded by the LincGlobal Program (LINCG23039), CSIC.

Data availability All of the data that support the findings of this study are available in the main text and in the Electronic supplementary material.

Declarations

Conflict of interest The authors have declared that no competing interests exist.

Ethical approval No ethical statement was reported.

Open Access This article is licensed under a Creative Commons Attribution 4.0 International License, which permits use, sharing, adaptation, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if changes were made. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit http://creativecommons.org/licenses/by/4.0/.

References

Álvarez MD (2001) Could peace be worse than war for Colombia's forests? Environmentalist 21:305–315. https://doi.org/10.1023/A:1012904318418

Bachman S, Moat J, Hill AW, la De Torre J, Scott B (2011) Supporting red list threat assessments with GeoCAT: geospatial conservation assessment tool. ZooKeys 150:117–126. https://doi.org/10.3897/zookeys.150.2109

Bernal R, Gradstein S, Celis M (2016) Catálogo de plantas y líquenes de Colombia, ver. 1.1. Instituto de Ciencias Naturales, Universidad Nacional de Colombia, Bogotá

Berry PE, Hipp AL, Wurdack KJ, Van Ee B, Riina R (2005) Molecular phylogenetics of the giant genus *Croton* and tribe Crotoneae (Euphorbiaceae sensu stricto) using ITS and *trnL-trnF* DNA sequence data. Amer J Bot 92:1520–1534. https://doi.org/10.3732/ajb.92.9.1520

Caruzo MBR, Riina R, Cordeiro I, Berry PE (2008) *Croton campanulatus* (Euphorbiaceae s.s.), a new species from the Brazilian Atlantic rain forest. Brittonia 60:261–264. https://doi.org/10.1007/s12228-008-9027-y

Chaves-Bedoya G (2022) Study on *Croton* sp. genetic diversity in the department of Norte de Santander using the internal transcribed spacer (ITS) region of ribosomal DNA (rDNA). Revista Colomb Ci Hortic 16:e13592. https://doi.org/10.17584/rcch.2022v16i1.13592

Cornejo X (2009) Two new species of *Pentagonia* (Rubiaceae, Hippotideae) from Colombia and Ecuador. Novon 19:25–31. https://doi.org/10.3417/2006060

Cornejo X, Mori SA (2012) *Grias angustipetala* and *G. ecuadorica*, two new species of Lecythidaceae from western Ecuador. Brittonia 64:318–324. https://doi.org/10.1007/s12228-012-9248-y

Elbers J (2011) *Las áreas protegidas de América Latina: Situación actual y perspectivas para el futuro*. Unión Internacional para la Conservación de la Naturaleza y de los Recursos Naturales (UICN), Quito

Farias SQ, Medeiros D, Riina R (2019) A new species of dragon's blood *Croton* (Euphorbiaceae) from Serra dos Órgãos (Rio de Janeiro, Brazil). PhytoKeys 126:13–24. https://doi.org/10.3897/phytokeys.126.35649

Feio AC, Ore-Rengifo MI, Berry PE, Riina R (2018a) Four new species of dragon's blood *Croton* (Euphorbiaceae) from South America. Syst Bot 43:212–220. https://doi.org/10.1600/036364418X697111

Feio AC, Meira RM, Riina R (2018b) Leaf anatomical features and their implications for the systematics of dragon's blood, *Croton* section *Cyclostigma* (Euphorbiaceae). Bot J Linn Soc 187:614–632. https://doi.org/10.1093/botlinnean/boy038

Feliner GN, Rosselló JA (2007) Better the devil you know? Guidelines for insightful utilization of nrDNA ITS in species-level evolutionary studies in plants. Molec Phylogen Evol 44:911–919. https://doi.org/10.1016/j.ympev.2007.01.013

Haber EA, Kainulainen K, Van Ee BW, Oyserman BO, Berry PE (2017) Phylogenetic relationships of a major diversification of *Croton* (Euphorbiaceae) in the western Indian Ocean region. Bot J Linn Soc 183:532–544. https://doi.org/10.1093/botlinnean/box004

IUCN Standards and Petitions Committee (2022) Guidelines for Using the IUCN Red List Categories and Criteria. Version 15.1. International Union for Conservation of Nature and Natural Resources

Jones K (2003) Review of Sangre de Drago (*Croton lechleri*) –a South American tree sap in the treatment of diarrhea, inflammation, insect bites, viral infections, and wounds: traditional uses to clinical research. J Altern Complement Med 9:877–896. https://doi.org/10.1089/107555303771952235

King S, Lázaro-Martín M, Fonseca R, Fonseca MP, Lozano-Díaz CG, García-Valles K (2020) Sustainable harvesting of Dragon's Blood *Croton lechleri* in Peru. HerbalGram 125:44–56

Lam-Tung N, Schmidt HA, von Haeseler A, Quang Minh B (2015) IQ-TREE: a fast and effective stochastic algorithm for estimating maximum likelihood phylogenies. Molec Biol Evol 32:268–274. https://doi.org/10.1093/molbev/msu300

Martín-Muñoz J, Silva EC, Gonzáles P, Muñoz-Sánchez A, Cumbicus N, Goodwin Z, Meira RMSA, Riina R (2024) *Croton maranonensis*: a new species of Euphorbiaceae from the tropical inter-Andean dry valleys. Eur J Taxon 947:289–308. https://doi.org/10.5852/ejt.2024.947.2639

Masa-Iranzo I, Sanmartín I, Caruzo MBR, Riina R (2021) Skipping the dry diagonal: spatiotemporal evolution of *Croton* section *Cleodora* (Euphorbiaceae) in the Neotropics. Bot J Linn Soc 197:61–84. https://doi.org/10.1093/botlinnean/boab016

Mateo-Ramírez L, Riina R (2020) *Croton calcareus*: a new species of dragon's blood (Euphorbiaceae) from dry forest in the state of Chiapas, Mexico. Eur J Taxon 657:1–14. https://doi.org/10.5852/ejt.2020.657

Meza EN (1999) Cosecha de Sangre de Grado (*Croton* spp.) y factores que influyen en su abundancia. In: Meza EN (ed) Desarrollando nuestra diversidad cultural: "Sangre de Grado" y el reto de su producción en el Perú. Fondo Editorial, Universidad Nacional Mayor de San Marcos, Lima, pp 45–76

Ministerio de Educación del Ecuador (2021) Cartilla AWA Zona 1. Available at: https://educacionbilingue.gob.ec/wp-content/uploads/2021/01/CARTILLA-AWA-ZONA-1_compressed.pdf. Accessed 23 Jun 2025

Murillo-A J (1999) Composición y distribución del género *Croton* (Euphorbiaceae) en Colombia, con cuatro especies nuevas. Caldasia 21:141–166

Myers N, Mittermeier RA, Mittermeier CG, Da Fonseca GA, Kent J (2000) Biodiversity hotspots for conservation priorities. Nature 403:853–858

Orejuela JE (1999) Traditional productive systems of the Awá (Cuaiaquer) Indians of southwestern Colombia and neighboring Ecuador. In: Redford KH, Padoch C (eds) Conservation of neotropical forests. Columbia University Press, New York, pp 58–82

Orozco CI, Canal D (2011) *Cuatresia anomala* y *Cuatresia physalana* (Physaleae, Solanaceae): dos especies nuevas de Colombia y Ecuador. Caldasia 33:79–89

Pereira ADPN, Riina R, Valduga E, Caruzo MBR (2022) A new species of *Croton* (Euphorbiaceae) endemic to the Brazilian Pampa and its phylogenetic affinities. Pl Syst Evol 308:14. https://doi.org/10.1007/s00606-021-01803-x

Pinto-Silva NP, de Souza KF, Silva OLM, Vitarelli NC, Pereira APN, Soares DA, Sodré RC, Medeiros D, Caruzo MBR, Carneiro Torres DS, da Silva MJ, Meira RMSA, Riina R, Feio AC (2023) Trichomes in the megadiverse genus *Croton* (Euphorbiaceae): a revised classification, identification parameters and standardized terminology. Bot J Linn Soc 203:37–49. https://doi.org/10.1093/botlinnean/boad008

Rambaut A (2006) FigTree v1.4.4. Available at: http://tree.bio.ed.ac.uk/software/figtree. Accessed 22 Aug 2022

Rambaut A, Suchard MA, Xie D, Drummond AJ (2014) Tracer v1.7.2. Available at: https://beast.community/tracer. Accessed 17 Aug 2025

Riina R, Berry PE, Cornejo X (2007) A new species of *"sangre de drago"* (*Croton* sect. *Cyclostigma*, Euphorbiaceae) from coastal Ecuador. Brittonia 59:97–101. https://doi.org/10.1663/0007-196X(2007)59[97:ANSOSD]2.0.CO;2

Riina R, Berry PE, van Ee BW (2009) Molecular phylogenetics of the dragon's blood *Croton* section *Cyclostigma* (Euphorbiaceae): a polyphyletic assemblage unraveled. Syst Bot 34:360–374. https://doi.org/10.1600/036364409788606415

Riina R, Carneiro-Torres DS, Peirson JA, Berry PE, Cordeiro I (2014) Further support for the Crotoneae phylogeny: a new species of *Brasiliocroton* (Euphorbiaceae) based on morphological, geographical, and molecular evidence. Syst Bot 39:227–234. https://doi.org/10.1600/036364414X678170

Riina R, van Ee BW, Caruzo MBR, Santos Carneiro-Torres D, dos Freitas Santos R, Berry PE (2021) The Neotropical *Croton sect. Geiseleria* (Euphorbiaceae): classification update, phylogenetic framework, and seven new species from South America. Ann Missouri Bot Gard 106:111–166. https://doi.org/10.3417/2021669

Ronquist F, Teslenko M, Van der Mark P, Ayres DL, Darling A, Höhna S, Larget B, Liu L, Suchard MA, Huelsenbeck JP (2012) MrBayes 3.2: efficient Bayesian phylogenetic inference and model choice across a large model space. Syst Biol 61:539–542. https://doi.org/10.1093/sysbio/sys029

Rossine Y, de Paula Souza Gomes A, de Laurênio Melo A, Athiê-Souza SM, de Ferreira Sales M (2020) *Croton suassunae* (Euphorbiaceae): a new dioecious species from Northeastern Brazil. Syst Bot 45:249–253. https://doi.org/10.1600/036364420X15862837791366

Rossine Y, Riina R, Silva OL, Louzada R (2025) Untangling the taxonomic knot of *Croton anomalus* (Euphorbiaceae), a Neotropical dry forest shrub. PeerJ 13:p.e19176. https://doi.org/10.7717/peerj.19176

Santos RF, Meira RMSA, Pereira ADPN, Miranda JDB, Caruzo MBR (2024) A new species of *Croton* section *Cleodora* (Euphorbiaceae ss) from Parque Estadual do Cantão, Tocantins, Brazil. Pl Syst Evol 310:32. https://doi.org/10.1007/s00606-024-01915-0

Santos RF, Riina R, Caruzo MBR (2017) Diversity of arborescent lineages of Crotoneae (Euphorbiaceae) in the Brazilian Atlantic Rain Forest. Pl Syst Evol 303:1467–1497. https://doi.org/10.1007/s00606-017-1441-8

Sierra CA, Mahecha M, Poveda G, Álvarez-Dávila E, Gutierrez-Velez VH, Reu B, Feilhauer H, Anáya J, Armenteras D, Benavides AM, Buendia C (2017) Monitoring ecological change during rapid socio-economic and political transitions: Colombian ecosystems in the post-conflict era. Environm Sci Policy 76:40–49. https://doi.org/10.1016/j.envsci.2017.06.011

Sodré RC, Da Silva MJ (2022) *Croton sertanejus*, a new species from Seasonally Dry Tropical Forest in Brazil, and redescription of *C. echioides* (Euphorbiaceae). Eur J Taxon 839:14–38. https://doi.org/10.5852/ejt.2022.839.1929

van Ee BW, Berry PE (2011) *Croton* section *Pedicellati* (Euphorbiaceae), a novel New World group, and a new subsectional classification of *Croton* section *Lamprocroton*. Syst Bot 36:88–98. https://doi.org/10.1600/036364411X553162

van Ee BW, Berry PE (2021) Taxonomy of *Croton flavens* (Euphorbiaceae), a Caribbean endemic, and a preliminary subsectional classification of *Croton* section *Adenophylli*. Caribb J Sci 51:314–357. https://doi.org/10.18475/cjos.v51i2.a14

van Ee BW, Riina R, Berry PE (2011) A revised synopsis and molecular phylogeny of the New World sections of *Croton* (Euphorbiaceae). Taxon 60:791–823. https://doi.org/10.1002/tax.603013

Vitarelli NC, Riina R, Caruzo MBR, Cordeiro I, Fuertes-Aguilar J, Meira RMSA (2015) Foliar secretory structures in Crotoneae (Euphorbiaceae): diversity, anatomy, and evolutionary significance. Amer J Bot 102:833–847. https://doi.org/10.3732/ajb.1500017

Zapata-Correa DA, Paz-López CA, Torrejano-Munevar AF, Berrío-Montoya S (2024) Una especie nueva de *Croton* sección *Geiseleria* (Euphorbiaceae) del cañón del río Cauca (Antioquia), noroccidente de Colombia. Darwiniana 12:284–294. https://doi.org/10.14522/darwiniana.2024.122.1246

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.